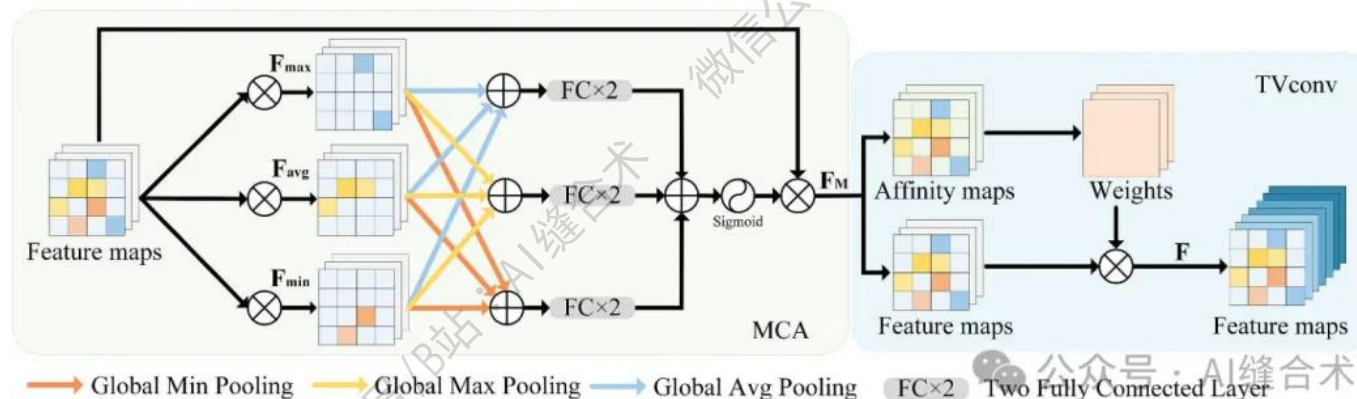


核心速览



论文题目: ELTDF-Net: a PAUT welding defect detection model based on locally perceptual residual gating and multi-frequency convolutional attention

中文题目: ELTDF -Net: 一种基于局部感知残差门控与多频卷积注意力的 PAUT 焊接缺陷检测模型

论文链接:

<https://doi.org/10.1080/10589759.2026.2619918>

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核心速览:

本文提出 ELTDF-Net 智能检测模型，通过三个创新模块（**ECCR 多通道感知局部特征**、**LCRB 残差门控机制**、**TVconv-CA 多频域卷积注意力**）增强焊接缺陷特征，解决 PAUT 检测中的噪声干扰、弱光特征及晶粒边界散射与缺陷频谱重叠问题，并开发智能分类软件系统验证工程实用性。

https://mp.weixin.qq.com/s/cCSBQJnQJKj3Yw5AV9Pn_w

```
MultiFrequencyChannelAttention(
  (fc): Sequential(
    (0): Conv2d(16, 1, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (1): ReLU(inplace=True)
    (2): Conv2d(1, 16, kernel_size=(1, 1), stride=(1, 1), bias=False)
  )
)
```

```
(average_channel_pooling): AdaptiveAvgPool2d(output_size=1)
(max_channel_pooling): AdaptiveMaxPool2d(output_size=1)
(model): TVConv(
  (weight_layers): Sequential(
    (0): _ConvBlock(
      (0): Conv2d(4, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (1): LayerNorm((64, 32, 32), eps=1e-05, elementwise_affine=True)
      (2): ReLU(inplace=True)
    )
    (1): _ConvBlock(
      (0): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (1): LayerNorm((64, 32, 32), eps=1e-05, elementwise_affine=True)
      (2): ReLU(inplace=True)
    )
    (2): _ConvBlock(
      (0): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (1): LayerNorm((64, 32, 32), eps=1e-05, elementwise_affine=True)
      (2): ReLU(inplace=True)
    )
    (3): _ConvBlock(
      (0): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
      (1): LayerNorm((64, 32, 32), eps=1e-05, elementwise_affine=True)
      (2): ReLU(inplace=True)
    )
    (4): Conv2d(64, 144, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
  )
  (unfold): Unfold(kernel_size=3, dilation=1, padding=1, stride=1)
)
input_tensor_shape : torch.Size([2, 16, 32, 32])
output_tensor_shape : torch.Size([2, 16, 32, 32])
```